

Domain Controller and Active Directory Lab

Part 3 – Active Directory Domain Services (AD DS) Installation and Domain Controller Promotion

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Technologies Used

- Oracle VirtualBox
- Windows Server 2022
- Server Manager
- Active Directory Domain Services

OBJECTIVE:

The objective of this lab is to install the Active Directory Domain Services (AD DS) role and promote the Windows Server 2022 system to a Domain Controller. This establishes the foundation for centralized authentication, directory services, DNS integration, and enterprise user management.

Lab Setup :

This lab continues from **Part 2 – Networking**, where the Windows Server 2022 virtual machine was configured with separate NAT and Internal Network adapters and assigned a static IPv4 address to support enterprise network services. With the networking infrastructure established, this phase focuses on installing the **Active Directory Domain Services (AD DS)** role through Server Manager and promoting the server to the first **Domain Controller** within a new Active Directory forest. Once promoted, the server provides centralized authentication, directory services, integrated DNS functionality, and the foundation for managing domain users, computers, and security policies throughout the enterprise environment.

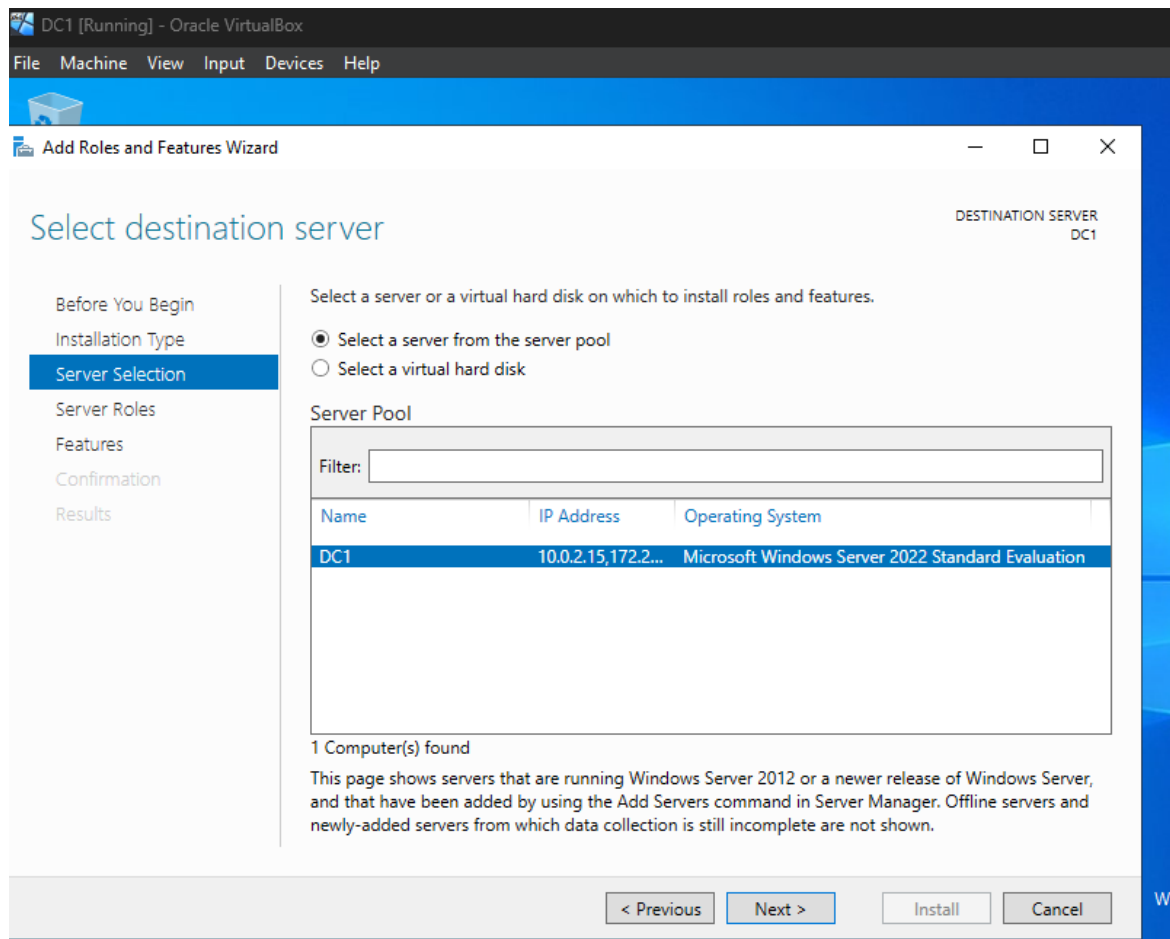


Figure 1. Selecting the Windows Server 2022 system as the destination server for the Active Directory Domain Services (AD DS) role installation.

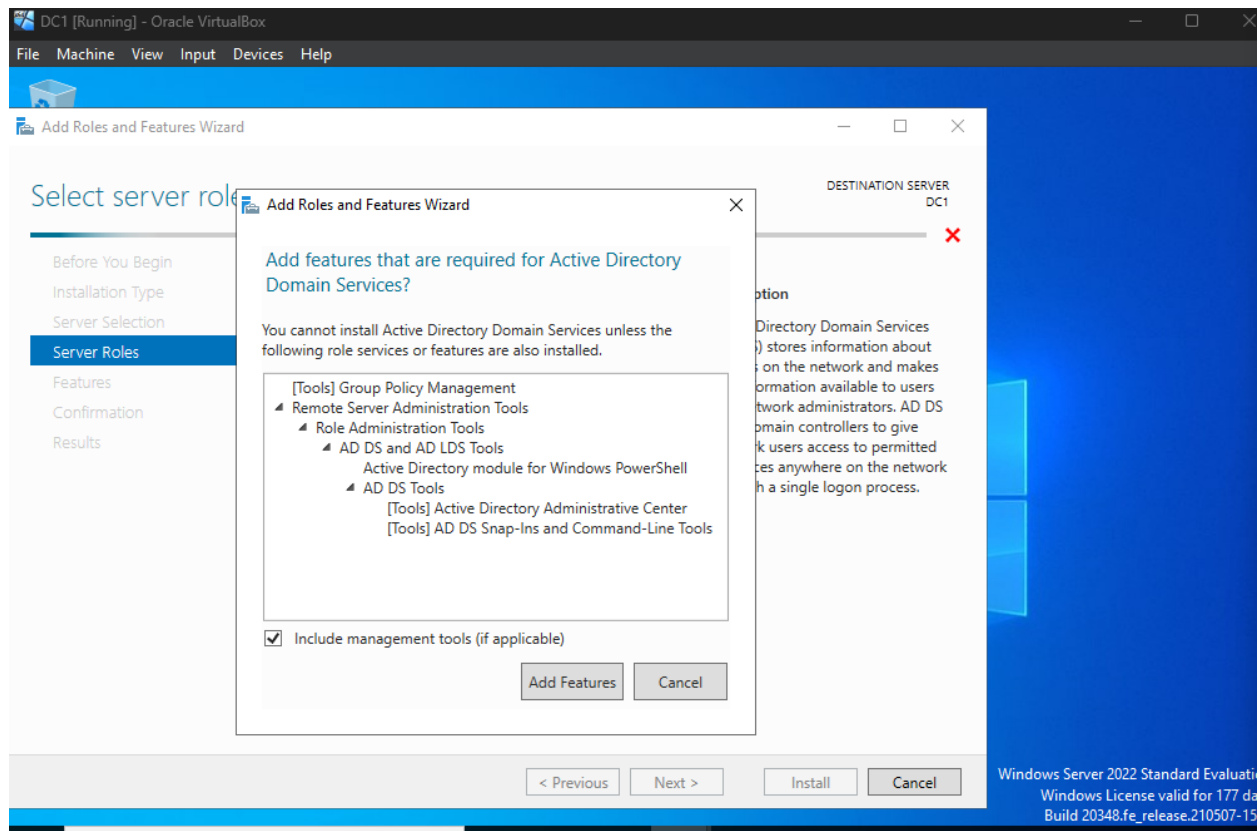


Figure 2. Installing the Active Directory Domain Services (AD DS) role and its required management tools on the Windows Server.

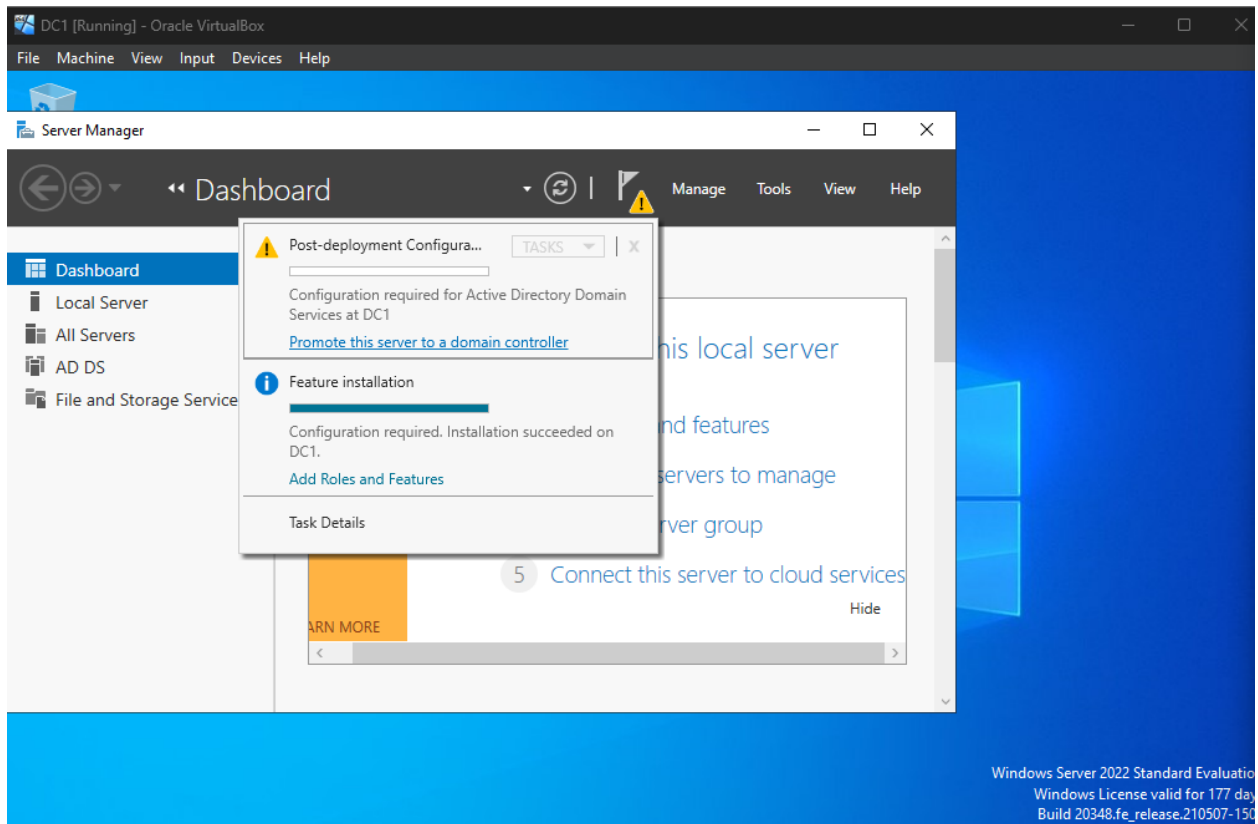


Figure 3. Beginning the Domain Controller promotion process by launching the Active Directory Domain Services Configuration Wizard.

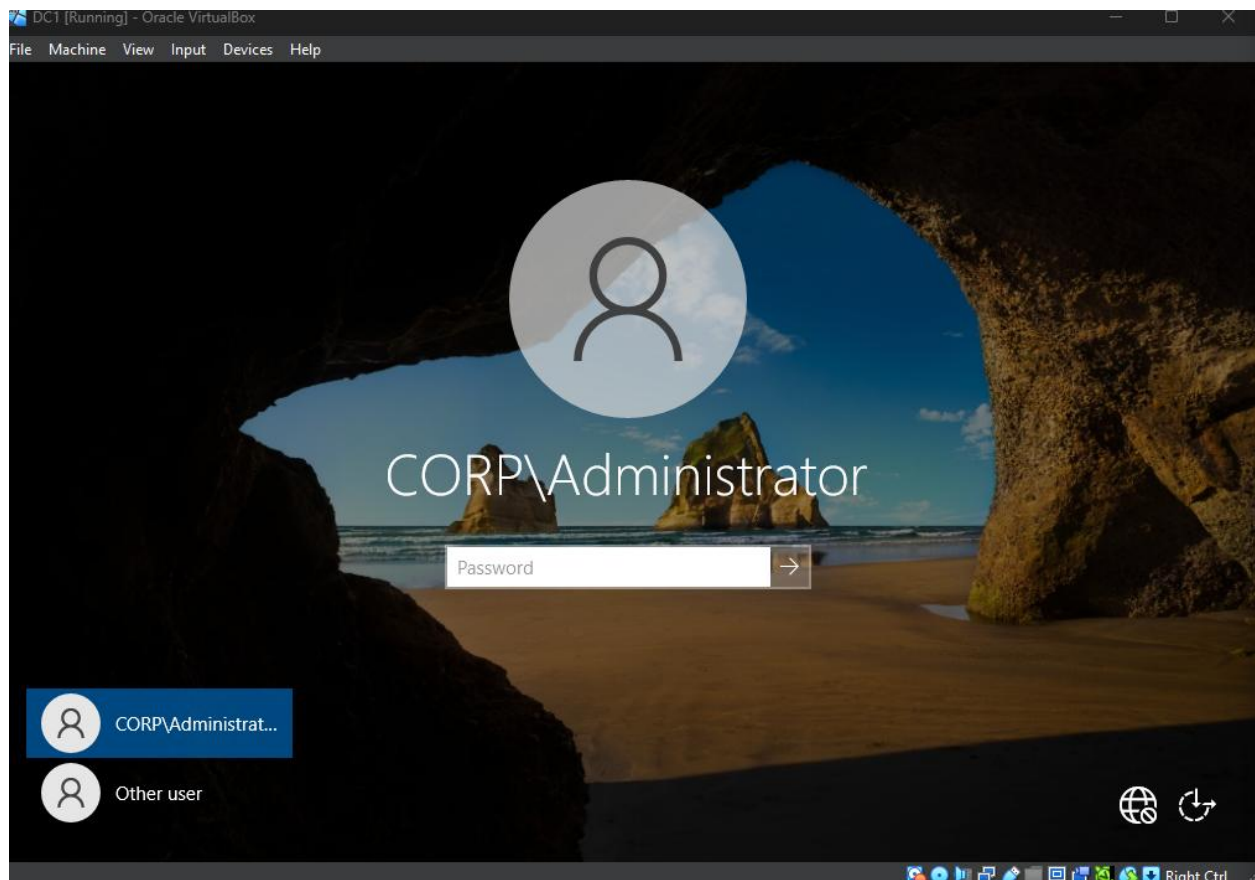


Figure 4. Successful promotion of the Windows Server to a Domain Controller. The server now authenticates users, hosts Active Directory Domain Services, and provides enterprise directory services for the domain.

What Changed?

Before promotion, the server functioned as a standard Windows Server installation. After promotion, it became the first Domain Controller within the new Active Directory forest. The server now hosts the Active Directory database, provides centralized authentication, integrates DNS services, and manages domain users, computers, and security policies.